

IBM Agentic AI Assignment

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Batch: Agentic AI – 04

1. Intent Classification

```
=====
Query: What are the pricing plans available for your software?
-----
Detected Intent : Sales
```

```
=====
Query: I forgot my account password.
-----
Detected Intent : Account
```

```
=====
Query: My application crashes whenever I upload a file.
-----
Detected Intent : Technical
```

```
=====
Query: I need a refund for my annual subscription.
-----
Detected Intent : Billing
```

```
=====
Query: What was my previous support issue?
-----
Detected Intent : Memory
```

2. Routing

```
=====
Query: What are the pricing plans available for your software?
-----
```

```
Detected Intent : Sales
```

```
Routed Department: Sales
```

```
=====
Query: I forgot my account password.
-----
```

```
Detected Intent : Account
```

```
Routed Department: Account
```

```
=====
Query: My application crashes whenever I upload a file.
-----
```

```
Detected Intent : Technical
```

```
Routed Department: Technical
```

```
=====
Query: I need a refund for my annual subscription.
-----
```

```
Detected Intent : Billing
```

```
Routed Department: Billing
```

```
=====
Query: What was my previous support issue?
-----
```

```
Detected Intent : Memory
```

```
Routed Department: Memory
```

3. Sales Agent

```
=====
Query: What are the pricing plans available for your software?
-----
```

```
Detected Intent : Sales
```

```
Routed Department: Sales
```

```
Retrieved Chunks :
```

```
[1] [Source: pricing_guide.txt]
```

```
# Pricing Guide
```

```
## Plans
```

```
We offer three main pricing plans:
```

1. Basic Plan: \$10/month or \$100/year. Includes 1 user, 10GB storage, and standard support (email only).
2. Professional Plan: \$25/month or \$250/year. Includes 5 users, 100GB storage, and priority support (email and chat).
3. Enterprise Plan: \$100/month or \$1000/year. Includes unlimited users, 1TB storage, 24/7 phone support, and a dedicated account manager.

```
[2] [Source: pricing_guide.txt]
```

```
## Add-ons
```

- Extra Storage: \$5/month per additional 50GB.
- Extra User (Basic/Pro): \$5/month per user.

```
All subscriptions renew automatically unless cancelled before the billing date.
```

```
[3] [Source: faq.txt]
```

```
# Frequently Asked Questions (FAQ)
```

```
Q: How do I change my billing cycle from monthly to annual?
```

```
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.
```

```
Q: Do you offer a free trial?
```

```
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.
```

```
Approval Required: No
```

```
Approval Decision: N/A
```

```

Supervisor Output: We offer three main pricing plans:

1. **Basic Plan**: $10/month or $100/year. Includes 1 user, 10GB storage, and standard support (email only).
2. **Professional Plan**: $25/month or $250/year. Includes 5 users, 100GB storage, and priority support (email and chat).
3. **Enterprise Plan**: $100/month or $1000/year. Includes unlimited users, 1TB storage, 24/7 phone support, and a dedicated account manager.

Would you like more details on any of these plans?
Final Response : We offer three main pricing plans:

1. **Basic Plan**: $10/month or $100/year. Includes 1 user, 10GB storage, and standard support (email only).
2. **Professional Plan**: $25/month or $250/year. Includes 5 users, 100GB storage, and priority support (email and chat).
3. **Enterprise Plan**: $100/month or $1000/year. Includes unlimited users, 1TB storage, 24/7 phone support, and a dedicated account manager.

Would you like more details on any of these plans?
=====

2026-06-27 23:27:37,502 - nodes - INFO - Classified intent: Account
2026-06-27 23:27:37,513 - rag - INFO - RAG retrieved 3 chunks.

```

4. Account Agent

```

=====
Query: I forgot my account password.
-----
Detected Intent : Account
Routed Department: Account
Retrieved Chunks :
[1] [Source: technical_manual.txt]
## Login Failures
If you see a "500 Internal Server Error" during login, our authentication servers might be under heavy load. Please wait 5 minutes and try again.
If you see an "Invalid Credentials" error, use the "Forgot Password" link to reset your account.
[2] [Source: company_policy.txt]
## Escalations
Any request for compensation, or explicit requests to speak with a manager or escalate an issue, must be flagged for management approval.

## Account Policies
Accounts that are inactive for 12 months will be suspended.
To reactivate, users must contact the Account department.
Account closure requests require management approval.
[3] [Source: faq.txt]
# Frequently Asked Questions (FAQ)

Q: How do I change my billing cycle from monthly to annual?
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.

Q: Do you offer a free trial?
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.
Approval Required: No
Approval Decision: N/A

Supervisor Output: If you see an "Invalid Credentials" error message during login, use the "Forgot Password" link to reset your account. This will guide you through r
esetting your password using the email address associated with your account.

If you continue to experience issues, please wait 5 minutes and try again as a server load might be causing delays. If the problem persists, contact our support team
for further assistance.
Final Response : If you see an "Invalid Credentials" error message during login, use the "Forgot Password" link to reset your account. This will guide you through r
esetting your password using the email address associated with your account.

If you continue to experience issues, please wait 5 minutes and try again as a server load might be causing delays. If the problem persists, contact our support team
for further assistance.
=====

2026-06-27 23:28:01,555 - nodes - INFO - Classified intent: Technical
2026-06-27 23:28:01,563 - rag - INFO - RAG retrieved 3 chunks.

```

5. Technical Agent

```

=====
Query: My application crashes whenever I upload a file.
-----
Detected Intent : Technical
Routed Department: Technical
Retrieved Chunks :
[1] [Source: technical_manual.txt]
## File Upload Failures
If the application crashes whenever you upload a file, it is a known issue in version 2.0.5.
To resolve this, you must update to version 2.0.6 or newer.
If the issue persists, ensure the uploaded file size is under 50MB and in a supported format (.pdf, .jpg, .png).
[2] [Source: technical_manual.txt]
# Technical Manual

## Application Crashes
If the application crashes upon startup, users should:
1. Clear their local cache directory located at '~/app_cache/'.
2. Ensure they are running version 2.1.0 or newer.
3. Check their operating system updates.
[3] [Source: technical_manual.txt]
## Login Failures
If you see a "500 Internal Server Error" during login, our authentication servers might be under heavy load. Please wait 5 minutes and try again.
If you see an "Invalid Credentials" error, use the "Forgot Password" link to reset your account.
Approval Required: No
Approval Decision: N/A

```

```

Supervisor Output: I understand that your application crashes whenever you upload a file. According to our technical manual, this is a known issue in version 2.0.5. To resolve it, please update your application to the latest version (version 2.0.6 or newer).

If updating does not solve the problem, ensure that the uploaded file size is under 50MB and that it's in one of our supported formats (.pdf, .jpg, .png).

Please let me know if you need further assistance with these steps.
Final Response : I understand that your application crashes whenever you upload a file. According to our technical manual, this is a known issue in version 2.0.5. To resolve it, please update your application to the latest version (version 2.0.6 or newer).

If updating does not solve the problem, ensure that the uploaded file size is under 50MB and that it's in one of our supported formats (.pdf, .jpg, .png).

Please let me know if you need further assistance with these steps.
=====
2026-06-27 23:28:33,125 - nodes - INFO - Classified intent: Billing
2026-06-27 23:28:33,134 - rag - INFO - RAG retrieved 3 chunks.

```

6. Billing Agent

```

=====
Query: I need a refund for my annual subscription.
=====
Detected Intent : Billing
Routed Department: Billing
Retrieved Chunks :
[1] [Source: company_policy.txt]
## Refunds and Cancellations
Customers are eligible for a full refund within 30 days of their initial purchase for any reason.
After 30 days, refunds are only issued on a pro-rated basis for annual subscriptions, and only if the service has experienced significant downtime (more than 48 hours in a given month) that was our fault.
All refund requests must be routed to the Billing department and must be approved by management.
[2] [Source: company_policy.txt]
Subscription cancellations take effect at the end of the current billing cycle. To cancel, a user must request it, which triggers a management review process.
[3] [Source: faq.txt]
# Frequently Asked Questions (FAQ)

Q: How do I change my billing cycle from monthly to annual?
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.

Q: Do you offer a free trial?
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.
Approval Required: Yes
Approval Decision: Approved

Supervisor Output: Sure, I can help with that. For an annual subscription, if you are eligible for a full refund within 30 days of your initial purchase due to any reason, we will process it according to our policy.

However, after the first 30 days, refunds on annual subscriptions are only issued on a pro-rated basis if there has been significant downtime that was our fault and exceeds more than 48 hours in a given month.

If you believe your situation qualifies for a refund based on these conditions, please submit a refund request through our support system. It will then be reviewed by management.

However, after the first 30 days, refunds on annual subscriptions are only issued on a pro-rated basis if there has been significant downtime that was our fault and exceeds more than 48 hours in a given month.

If you believe your situation qualifies for a refund based on these conditions, please submit a refund request through our support system. It will then be reviewed by management.

Please note that all refund requests must go through the Billing department and require approval from management before any action is taken.
=====
2026-06-27 23:29:21,549 - nodes - INFO - Classified intent: Memory
2026-06-27 23:29:21,551 - rag - INFO - RAG skipped: Memory intent query.
2026-06-27 23:29:24,044 - memory - INFO - Memory recall: retrieved 1 records for David.

```

7. RAG Retrieval

```

Retrieved Chunks :
[1] [Source: pricing_guide.txt]
# Pricing Guide

## Plans
We offer three main pricing plans:

1. Basic Plan: $10/month or $100/year. Includes 1 user, 10GB storage, and standard support (email only).
2. Professional Plan: $25/month or $250/year. Includes 5 users, 100GB storage, and priority support (email and chat).
3. Enterprise Plan: $100/month or $1000/year. Includes unlimited users, 1TB storage, 24/7 phone support, and a dedicated account manager.
[2] [Source: pricing_guide.txt]
## Add-ons
- Extra Storage: $5/month per additional 50GB.
- Extra User (Basic/Pro): $5/month per user.

All subscriptions renew automatically unless cancelled before the billing date.
[3] [Source: faq.txt]
# Frequently Asked Questions (FAQ)

Q: How do I change my billing cycle from monthly to annual?
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.

Q: Do you offer a free trial?
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.

```

```

Retrieved Chunks :
[1] [Source: technical_manual.txt]
## Login Failures
If you see a "500 Internal Server Error" during login, our authentication servers might be under heavy load. Please wait 5 minutes and try again.
If you see an "Invalid Credentials" error, use the "Forgot Password" link to reset your account.
[2] [Source: company_policy.txt]
## Escalations
Any request for compensation, or explicit requests to speak with a manager or escalate an issue, must be flagged for management approval.

## Account Policies
Accounts that are inactive for 12 months will be suspended.
To reactivate, users must contact the Account department.
Account closure requests require management approval.
[3] [Source: faq.txt]
# Frequently Asked Questions (FAQ)

Q: How do I change my billing cycle from monthly to annual?
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.

Q: Do you offer a free trial?
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.

```

```

Retrieved Chunks :
[1] [Source: technical_manual.txt]
## File Upload Failures
If the application crashes whenever you upload a file, it is a known issue in version 2.0.5.
To resolve this, you must update to version 2.0.6 or newer.
If the issue persists, ensure the uploaded file size is under 50MB and in a supported format (.pdf, .jpg, .png).
[2] [Source: technical_manual.txt]
# Technical Manual

## Application Crashes
If the application crashes upon startup, users should:
1. Clear their local cache directory located at `~/.app_cache/`.
2. Ensure they are running version 2.1.0 or newer.
3. Check their operating system updates.
[3] [Source: technical_manual.txt]
## Login Failures
If you see a "500 Internal Server Error" during login, our authentication servers might be under heavy load. Please wait 5 minutes and try again.
If you see an "Invalid Credentials" error, use the "Forgot Password" link to reset your account.

```

```

Retrieved Chunks :
[1] [Source: company_policy.txt]
## Refunds and Cancellations
Customers are eligible for a full refund within 30 days of their initial purchase for any reason.
After 30 days, refunds are only issued on a pro-rated basis for annual subscriptions, and only if the service has experienced significant downtime (more than 48 hours in a given month) that was our fault.
All refund requests must be routed to the Billing department and must be approved by management.
[2] [Source: company_policy.txt]
Subscription cancellations take effect at the end of the current billing cycle. To cancel, a user must request it, which triggers a management review process.
[3] [Source: faq.txt]
# Frequently Asked Questions (FAQ)

Q: How do I change my billing cycle from monthly to annual?
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.

Q: Do you offer a free trial?
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.

```

8. Human Approval

```

[HUMAN APPROVAL REQUIRED]
Customer: David
Request: I need a refund for my annual subscription.
Draft Response: Sure, I can help with that. For an annual subscription, if you are eligible for a full refund within 30 days of your initial purchase due to any reason, we will process it according to our policy.

However, after the first 30 days, refunds on annual subscriptions are only issued on a pro-rated basis if there has been significant downtime that was our fault and exceeds more than 48 hours in a given month.

If you believe your situation qualifies for a refund based on these conditions, please submit a refund request through our support system. It will then be reviewed by management.

Please note that all refund requests must go through the Billing department and require approval from management before any action is taken.

Approve? (Y/N): y
2026-06-27 23:28:59,263 - human_loop - INFO - Human approved the response.

```

9. Memory Recall

```

=====
Query: I need a refund for my annual subscription.
-----
Detected Intent : Billing
Routed Department: Billing
Retrieved Chunks :
  [1] [Source: company_policy.txt]
## Refunds and Cancellations
Customers are eligible for a full refund within 30 days of their initial purchase for any reason.
After 30 days, refunds are only issued on a pro-rated basis for annual subscriptions, and only if the service has experienced significant downtime (more than 48 hours
in a given month) that was our fault.
All refund requests must be routed to the Billing department and must be approved by management.
  [2] [Source: company_policy.txt]
Subscription cancellations take effect at the end of the current billing cycle. To cancel, a user must request it, which triggers a management review process.
  [3] [Source: faq.txt]
# Frequently Asked Questions (FAQ)

Q: How do I change my billing cycle from monthly to annual?
A: You can change your billing cycle in the Account Settings -> Billing tab. The change will take effect immediately and you will be charged a prorated amount.

Q: Do you offer a free trial?
A: Yes, we offer a 14-day free trial for the Professional Plan. No credit card is required.
Approval Required: Yes
Approval Decision: Approved
Supervisor Output: Sure, I can help with that. For an annual subscription, if you are eligible for a full refund within 30 days of your initial purchase due to any re
ason, we will process it according to our policy.

However, after the first 30 days, refunds on annual subscriptions are only issued on a pro-rated basis if there has been significant downtime that was our fault and e
xceeds more than 48 hours in a given month.

If you believe your situation qualifies for a refund based on these conditions, please submit a refund request through our support system. It will then be reviewed by
management.

```

```

2026-06-27 23:29:21,549 - nodes - INFO - Classified intent: Memory
2026-06-27 23:29:21,551 - rag - INFO - RAG skipped: Memory intent query.
2026-06-27 23:29:24,044 - memory - INFO - Memory recall: retrieved 1 records for David.

```

```

=====
Query: What was my previous support issue?
-----
Detected Intent : Memory
Routed Department: Memory
Retrieved Chunks :
  None
Approval Required: No
Approval Decision: N/A
Supervisor Output: Your previous support issue was about needing a refund for your annual subscription.
Final Response : Your previous support issue was about needing a refund for your annual subscription.
=====

```

10. Supervisor Agent

Supervisor Output: We offer three main pricing plans:

1. **Basic Plan**: \$10/month or \$100/year. Includes 1 user, 10GB storage, and standard support (email only).
2. **Professional Plan**: \$25/month or \$250/year. Includes 5 users, 100GB storage, and priority support (email and chat).
3. **Enterprise Plan**: \$100/month or \$1000/year. Includes unlimited users, 1TB storage, 24/7 phone support, and a dedicated account manager.

Would you like more details on any of these plans?

Supervisor Output: If you see an "Invalid Credentials" error message during login, use the "Forgot Password" link to reset your account. This will guide you through r
esetting your password using the email address associated with your account.

If you continue to experience issues, please wait 5 minutes and try again as a server load might be causing delays. If the problem persists, contact our support team
for further assistance.

Supervisor Output: I understand that your application crashes whenever you upload a file. According to our technical manual, this is a known issue in version 2.0.5. T
o resolve it, please update your application to the latest version (version 2.0.6 or newer).

If updating does not solve the problem, ensure that the uploaded file size is under 50MB and that it's in one of our supported formats (.pdf, .jpg, .png).

Please let me know if you need further assistance with these steps.

Supervisor Output: Sure, I can help with that. For an annual subscription, if you are eligible for a full refund within 30 days of your initial purchase due to any re
ason, we will process it according to our policy.

However, after the first 30 days, refunds on annual subscriptions are only issued on a pro-rated basis if there has been significant downtime that was our fault and e
xceeds more than 48 hours in a given month.

If you believe your situation qualifies for a refund based on these conditions, please submit a refund request through our support system. It will then be reviewed by
management.

11. SQLite DB

		customer...	session_id	user_query	assistant_response	timestamp
		Filter...	Filter...	Filter...	Filter...	Filter...
1	1	David	2d581bf-074f-420b-84ba-d2a0297ea983	What are the pricing plans available for your software?	1. Basic Plan: \$10/month or \$100/year. Includes 1 user, 10...	2026-06-27 14:35:04.89...
2	2	David	2d581bf-074f-420b-84ba-d2a0297ea983	I forgot my account password.	Okay, I understand you've forgotten your account passwo...	2026-06-27 14:39:36.07...
3	3	David	2d581bf-074f-420b-84ba-d2a0297ea983	My application crashes whenever I upload a file.	I understand your frustration with this issue. Let's troubles...	2026-06-27 14:45:16.06...
4	4	David	2d581bf-074f-420b-84ba-d2a0297ea983	I need a refund for my annual subscription.	Thank you for contacting us regarding your refund reques...	2026-06-27 14:53:06.28...
5	5	David	2d581bf-074f-420b-84ba-d2a0297ea983	What was my previous support issue?	Could you please provide the account name or email addr...	2026-06-27 14:54:14.61...
6	6	David	9b064d0-a2fa-4c42-8855-9644d90b382b	What are the pricing plans available for your software?	We offer three main pricing plans: 1. Basic plan: \$10/mont...	2026-06-27 15:17:59.41...
7	7	David	9b064d0-a2fa-4c42-8855-9644d90b382b	I forgot my account password.	Okay, I understand you've forgotten your account passwo...	2026-06-27 15:18:20.35...

12. Knowledge Base

- company_policy.txt
- faq.txt
- pricing_guide.txt
- technical_manual.txt